

Joint Calibration: Reproducing Roughness, the Leverage Effect, the Taylor Effect, the Zumbach Effect, and Excess Kurtosis Simultaneously, Per Asset

Companion note to the ESN A2-981003 market-data hyperparameter search

September 5, 2026

Abstract

Every note in this family so far has calibrated the ESN–A2 architecture to *one* stylised fact at a time. This note calibrates all 5 – roughness (the moment-based Hurst exponent), the leverage effect, the Taylor effect, the Zumbach effect, and excess kurtosis – simultaneously, per asset, via a single combined objective and a 6-parameter search $\theta = (H, \text{rough_scale}, \lambda_{lo}, \lambda_{hi}, m_1, \alpha)$. α is a feedback layer: the raw pre-activation variance feeds an EWMA recursion producing a 2nd, smoothed signal that recursively depends on its own previous value, used to generate returns (giving Taylor and kurtosis genuine multi-day persistence) while Hurst, leverage, and Zumbach all read the unsmoothed signal directly. Each lag-indexed statistic is additionally smoothed with a light 3-point moving average across adjacent lags before entering the residual, applied identically to the ESN’s own simulated curve and the real target curve; Taylor’s own 2 curves are fit separately (not their difference) and Zumbach on its own raw level (not shape-normalised), neither carrying an extra weight multiplier. A 2nd, independent fix corrects the optimiser itself: freezing the random seed within each calibration run (common random numbers) produced clean, monotonic convergence for every asset. Combined, roughness and kurtosis are essentially perfect (9/9), leverage remains excellent, and – the central result of this note – the Taylor effect, which collapsed to a strict 0% floor without the feedback layer, recovers to 33–100% coverage for 8 of 9 assets; the Zumbach effect’s own qualitative shape (rising vs. falling with horizon) is also substantially more often correct. **RUSSELL2000** is the one clear exception, regressing on Taylor and Zumbach under this combination.

Contents

1	Motivation and set-up	2
2	The model	2
2.1	The reservoir	2
2.2	The z -bank	2
2.3	Pre-activation and the feedback layer	3
3	Methodology	4
3.1	The calibrated parameter vector	4
3.2	Smoothed statistics	4
3.3	Taylor fit as 2 separate curves; Zumbach fit on its own raw level	4
3.4	The combined residual vector	4
3.4.1	The 6 blocks	4

3.4.2	Per-block normalisation	4
3.4.3	Extra weight multipliers and leverage's own lag weighting	5
3.4.4	The full residual, written out	5
3.5	Common random numbers	6
3.6	Per-asset starting points	6
3.7	Stage 2: kurtosis-targeted bisection for m_1	6
4	Results	6
5	Limitations and Recommended Next Steps	14

1 Motivation and set-up

The 9-asset bench is the same throughout this note family: 6 stocks (LHX, BK, HUM, TSN, IEX, KMB) and 3 indices (SP500, RUSSELL2000, NASDAQ100). Real targets for leverage, Taylor, Zumbach, and kurtosis are taken directly from those notes' own computations; the real, bias-corrected moment-based Hurst exponent is computed fresh for this same bench. $N_r = 22, N_z = 29$ and the 7 shared z -bank hyperparameters are fixed at the companion note's own optimum throughout.

2 The model

The simulator is a 2-layer echo-state network: a linear reservoir of Ornstein–Uhlenbeck modes, and a nonlinear z -bank reading 2 taps off the reservoir, with a feedback-layer split of the resulting variance signal.

2.1 The reservoir

$r_t = (r_{t,1}, \dots, r_{t,N_r})$ is a bank of independent Ornstein–Uhlenbeck processes with geometrically spaced rates,

$$\lambda_i = \lambda_{lo} \left(\frac{\lambda_{hi}}{\lambda_{lo}} \right)^{\frac{i-1}{N_r-1}}, \quad i = 1, \dots, N_r, \quad (1)$$

updated once per day ($dt = 1$) from a single shared innovation $\varepsilon_t \sim \mathcal{N}(0, 1)$:

$$r_{t+1,i} = \alpha_i^{\text{OU}} r_{t,i} + c_i \varepsilon_t, \quad \alpha_i^{\text{OU}} = e^{-\lambda_i dt}, \quad c_i = \sqrt{1 - (\alpha_i^{\text{OU}})^2}, \quad (2)$$

(the reservoir's own per-mode decay α_i^{OU} is unrelated to the calibrated feedback parameter α of §2.3). The reservoir is read through a single fixed direction $q \in \mathbb{R}^{N_r}$, $q_i \propto \lambda_i^{0.5-H}$, unit-normalised and then variance-normalised against the reservoir's own stationary autocovariance so that $q^\top C q = 1$.

2.2 The z -bank

$z_t = (z_{t,1}, \dots, z_{t,N_z})$ is a nonlinear bank with its own geometrically spaced decay rates. Each unit reads 2 fixed reservoir taps, $p_1 = r_{t+1, \text{fast}(j)}$, $p_2 = r_{t+1, \text{slow}(j)}$:

$$z_{t+1,j} = e^{-a_j dt} z_{t,j} + (1 - e^{-a_j dt}) m_{t+1} \mathbf{z_strength} \tanh(u_{t,j}), \quad (3)$$

$$\begin{aligned} u_{t,j} = & s_j p_1 p_2 + \mathbf{even_strength}(0.7 p_1^2 + 0.3 p_2^2 - 1) - \mathbf{linear_strength}(0.7 p_1 + 0.3 p_2) \\ & + \mathbf{zz_scale}_j z_{t,j} + \frac{1}{2} \mathbf{local_z_strength}(z_{t,j-1} + z_{t,j+1}), \end{aligned} \quad (4)$$

with the gate $m_{t+1} = \max(0, 1 - \text{gamma_norm} \|r_{t+1}\|/\sqrt{N_r})$ suppressing the z -bank's own update whenever the reservoir's current norm is large.

2.3 Pre-activation and the feedback layer

Using r_t, z_t (the state before this step's own update),

$$\eta_t = b_0 + \rho_r \text{rough_scale}(q^\top r_t) + \sum_{j=1}^{N_z} w_{j,z} z_{j,t} + \frac{1}{N_r} \left(m_1 \|r_t\|^2 + m_2 (q^\top r_t)^2 \right), \quad (5)$$

m_2 held fixed at 0; only m_1 calibrated. The raw pre-activation variance is

$$(\sigma_t^{\text{raw}})^2 = \text{SIG_MIN}^2 + \text{softplus}(\eta_t)^2, \quad (6)$$

which feeds a genuine feedback layer – an EWMA recursion depending on its own previous value, not just on σ_t^{raw} :

$$(\sigma_t^{\text{used}})^2 = (1 - \alpha)(\sigma_{t-1}^{\text{used}})^2 + \alpha(\sigma_t^{\text{raw}})^2. \quad (7)$$

Daily log-returns are generated from σ_t^{used} , $dX_t = -\frac{1}{2}(\sigma_t^{\text{used}})^2 + \sigma_t^{\text{used}} \varepsilon_t$ (reusing the same ε_t that updates r_{t+1}). $\alpha = 1$ recovers $\sigma_t^{\text{used}} \equiv \sigma_t^{\text{raw}}$ exactly (verified bit-identical), i.e. a single, unsmoothed signal. Hurst, leverage, and Zumbach all read σ_t^{raw} directly (an EWMA is itself autocorrelated, and was found to inject a spurious dip-and-recovery shape into leverage's own $\text{corr}(x_t, v_{t+L})$ when read from a smoothed signal); Taylor and kurtosis, functions of returns alone, inherit whatever persistence the feedback layer injects into X_t .

Interpretation. σ_t^{raw} is the reservoir's own *instantaneous* read of volatility: a snapshot, at day t , of what the current configuration of OU modes and z -bank units implies about the width of that day's own return distribution, with no memory of its own past values beyond what the reservoir itself already carries. It is the right signal for the 3 stylised facts that are themselves defined directly on the instantaneous variance path – Hurst's own structure function characterises how $\log \sigma_t^{\text{raw}}$ scales with lag, and leverage and Zumbach both pair σ_t^{raw} against x_t at a lag; smoothing this signal before using it for any of the 3 would distort exactly the scaling or timing relationship each is meant to measure (§2.3's own leverage example). σ_t^{used} plays a different role: it is the market's own *realised, tradeable* volatility, the object that actually sets the scale of that day's return through dX_t . Because it is built by exponentially averaging σ_t^{raw} over its own recent history rather than reading it instantaneously, σ_t^{used} carries genuine memory that σ_t^{raw} alone does not – a large realisation does not vanish the instant the reservoir's own state moves on, but decays over the horizon set by α (a small α , long memory; $\alpha \rightarrow 1$, no memory beyond the instantaneous read). This is precisely the persistence the Taylor and Zumbach effects, both statements about how long volatility shocks take to fade from the return process itself, need to draw on: routing returns through σ_t^{used} rather than σ_t^{raw} gives the model a 2nd, independent timescale for memory in *returns* specifically, separate from whatever timescale λ_{hi} and λ_{lo} set for the reservoir's own roughness. Without this readout, the single shared signal forces 1 timescale to serve both roles: an earlier, single-signal version of this note found H pinning λ_{hi} high to keep $\hat{H}$ under control, leaving the return process itself with no room to carry the multi-day persistence Taylor and Zumbach require – the same conflict this paragraph's own 2nd timescale is built to avoid.

3 Methodology

3.1 The calibrated parameter vector

$$\begin{aligned}\theta &= (H, \text{rough_scale}, \lambda_{lo}, \lambda_{hi}, m_1, \alpha) \\ &\in [0.01, 2.5] \times [0.2, 0.7] \times [10^{-5}, 0.1] \times [0.001, 5.0] \times [-1.0, 0.5] \times [0.02, 1.0].\end{aligned}\tag{8}$$

3.2 Smoothed statistics

Each lag-indexed statistic – leverage, $\text{acf}(|x|)$, $\text{acf}(x^2)$, Zumbach’s own $Z(L)$ – is passed through a light 3-point moving average across adjacent lags (shrinking window at the edges, no wraparound) *before* entering the residual, applied identically to the ESN’s own simulated curve and the real target curve, so the comparison itself is unaffected by the smoothing’s own choice of window.

3.3 Taylor fit as 2 separate curves; Zumbach fit on its own raw level

$\text{acf}(|x|)$ and $\text{acf}(x^2)$ are each fit separately against their own (smoothed) real target; Zumbach is fit against its own raw (smoothed) $Z(L)$ – no shape normalisation. Neither carries an extra weight multiplier beyond the standard tolerance/block-count normalisation of §3.4.

3.4 The combined residual vector

One evaluation of θ simulates K independent paths of length T days, extracts all 5 stylised facts from each simulated path, and averages each statistic across the K paths before it ever enters the residual – so the residual compares an averaged simulated statistic against a single real value, not K separate comparisons. This subsection gives the exact construction, block by block.

3.4.1 The 6 blocks

The residual vector $r(\theta) \in \mathbb{R}^{53}$ concatenates 6 blocks, one per stylised fact (Taylor’s own 2 curves counted separately, per §3.3):

Block	Quantity compared	# terms	Lags/horizons used
Hhat	$\hat{H}_{\text{ESN}}$ vs. $\hat{H}_{\text{real}}$	1	–
leverage	$\text{corr}(x_t, \sigma_{t+L}^2)$	40	$L = 1, \dots, 40$
$\text{acf}(x)$	$\text{acf}(x_t , L)$	6	$L \in \{1, 3, 5, 10, 15, 20\}$
$\text{acf}(x^2)$	$\text{acf}(x_t^2, L)$	6	$L \in \{1, 3, 5, 10, 15, 20\}$
Zumbach	$Z(L)$	5	$L \in \{2, 10, 20, 30, 40\}$
kurtosis	excess kurtosis of x_t	1	–
Total		53	

Every lag-indexed quantity (leverage, both Taylor curves, Zumbach) is smoothed via a light 3-point moving average across adjacent lags (§3.2) *before* it enters any of the formulas below, on both the ESN’s own simulated side and the real target side.

3.4.2 Per-block normalisation

Each block’s own raw difference (ESN average minus real target) is divided by 2 factors before being placed into $r(\theta)$:

1. **A cross-asset tolerance** s_k – the standard deviation of that stylised fact’s own real value across the 9-asset bench, computed once and held fixed:

$$s_{\text{Hhat}} = 0.0166, \quad s_{\text{lev}} = 0.0247, \quad s_{\text{abs_acf}} = 0.0768, \quad s_{\text{sq_acf}} = 0.0894,$$

$$s_{\text{zum}} = 0.0807, \quad s_{\text{kurt}} = 3.3962.$$

Dividing by s_k puts every block’s own residual in comparable, roughly $O(1)$ -per-term units regardless of that fact’s own natural scale (a raw kurtosis residual is $O(1-10)$; a raw leverage residual is $O(0.01-0.1)$ – without this step, kurtosis alone would dominate the total sum of squares by roughly 2 orders of magnitude).

2. **$\sqrt{n_{\text{terms}}}$ for that block** – leverage’s own 40 terms would otherwise outvote Hhat’s own single term even with each term individually well-scaled by s_k , since the total sum of squares grows with term count. Dividing each term by $\sqrt{n_{\text{terms}}}$ means a block where every term sits at “1 tolerance unit” of error contributes the same total, $n_{\text{terms}} \times (1/\sqrt{n_{\text{terms}}})^2 = 1$, to $\|r(\theta)\|_2^2$ regardless of how many terms it has.

3.4.3 Extra weight multipliers and leverage’s own lag weighting

On top of the normalisation above, each block carries a fixed multiplier w_k : $w_{\text{Hhat}} = 3$, and $w_k = 1$ for every other block (leverage, both Taylor curves, Zumbach, kurtosis) – per direct request, Taylor and Zumbach carry no extra weight beyond §3.4.2’s own tolerance/count normalisation. Leverage’s own 40 terms are additionally weighted by lag, $u_L = 1/(L+1)^2$, giving the short lags (where the leverage shape carries the most reliable signal) far more influence than the long ones.

3.4.4 The full residual, written out

Writing $\overline{(\cdot)}$ for the K -path average of a simulated statistic and $(\cdot)^{\text{real}}$ for the corresponding real target, The residual vector concatenates the following 6 blocks, in order:

$$r_{\text{Hhat}} = \frac{w_{\text{Hhat}}(\overline{\hat{H}} - \hat{H}^{\text{real}})}{s_{\text{Hhat}}\sqrt{1}} \quad (1 \text{ term}) \quad (9)$$

$$r_{\text{lev}}(L) = \frac{w_{\text{lev}} u_L (\overline{\text{lev}}(L) - \text{lev}^{\text{real}}(L))}{s_{\text{lev}}\sqrt{40}}, \quad L = 1, \dots, 40 \quad (40 \text{ terms}) \quad (10)$$

$$r_{\text{abs}}(L) = \frac{\overline{\text{acf}}(|x|, L) - \text{acf}^{\text{real}}(|x|, L)}{s_{\text{abs_acf}}\sqrt{6}}, \quad L \in \{1, 3, 5, 10, 15, 20\} \quad (6 \text{ terms}) \quad (11)$$

$$r_{\text{sq}}(L) = \frac{\overline{\text{acf}}(x^2, L) - \text{acf}^{\text{real}}(x^2, L)}{s_{\text{sq_acf}}\sqrt{6}}, \quad L \in \{1, 3, 5, 10, 15, 20\} \quad (6 \text{ terms}) \quad (12)$$

$$r_{\text{zum}}(L) = \frac{\overline{Z}(L) - Z^{\text{real}}(L)}{s_{\text{zum}}\sqrt{5}}, \quad L \in \{2, 10, 20, 30, 40\} \quad (5 \text{ terms}) \quad (13)$$

$$r_{\text{kurt}} = \frac{\overline{\text{kurt}} - \text{kurt}^{\text{real}}}{s_{\text{kurt}}\sqrt{1}} \quad (1 \text{ term}) \quad (14)$$

$$r(\theta) = (r_{\text{Hhat}}, r_{\text{lev}}(1), \dots, r_{\text{lev}}(40), r_{\text{abs}}(1), \dots, r_{\text{abs}}(20), r_{\text{sq}}(1), \dots, r_{\text{sq}}(20), r_{\text{zum}}(2), \dots, r_{\text{zum}}(40), r_{\text{kurt}}) \in \mathbb{R}^{53} \quad (15)$$

with $w_{\text{Hhat}} = 3$, $w_{\text{lev}} = 1$, $u_L = (L+1)^{-2}$. Bounded Levenberg–Marquardt (`scipy.optimize.least_squares`, `method="trf"`, bounds Eq. 8) minimises $\|r(\theta)\|_2^2$, the sum of squares of all 53 entries.

3.5 Common random numbers

scipy’s own finite-difference Jacobian estimate probes $\theta + d\theta \cdot e_i$ for each parameter i . Drawing an independent random seed at every such probe was found to make the estimated gradient dominated by sampling noise rather than the residual’s own true local sensitivity. Freezing the random seed within 1 `calibrate_joint` call means every probe reuses the SAME underlying random paths as the base point, so most of the sampling noise cancels in the difference, leaving the genuine local gradient.

3.6 Per-asset starting points

Stage-1 searches were started from the H -to- $\hat{H}$ map established earlier in this note family, inverted against each asset’s own real Hurst target to give a precise starting H_0 ; $\alpha_0 = 0.1$ throughout.

3.7 Stage 2: kurtosis-targeted bisection for m_1

Bounded bisection (`scipy.optimize.brentq`), holding $H, \text{rough_scale}, \lambda_{lo}, \lambda_{hi}, \alpha$ fixed at their own stage-1 values and targeting kurtosis specifically, with heavy, fixed-seed averaging (24 paths, $T = 4,000$ days), resolves the noisy-gradient difficulty m_1 alone faces in the joint search.

4 Results

Table 1: Final calibrated parameters, all 9 assets.

Asset	type	H	<code>rough_scale</code>	λ_{lo}	λ_{hi}	m_1	α
LHX	stock	0.070	0.436	0.00192	4.9779	+0.1564	0.476
BK	stock	0.146	0.544	0.00103	4.1920	+0.2551	0.138
HUM	stock	0.104	0.446	0.00107	4.7868	+0.1623	0.492
TSN	stock	0.013	0.592	0.00003	4.9971	+0.1632	0.483
IEX	stock	0.010	0.379	0.00004	4.9904	+0.2220	0.370
KMB	stock	0.211	0.671	0.00001	4.9471	+0.0483	0.570
SP500	index	0.386	0.672	0.00001	4.8809	+0.1745	0.220
RUSSELL2000	index	0.267	0.541	0.00336	4.4327	+0.0870	0.229
NASDAQ100	index	0.322	0.700	0.00001	5.0000	+0.0958	0.128

Every asset’s own α lands in $[0.13, 0.57]$, genuinely using the feedback layer everywhere rather than collapsing to 1 (no smoothing); λ_{hi} still sits at or near its own upper bound for every asset, as in the pure single-signal version, but the feedback layer now gives Taylor and Zumbach a source of persistence that does not require λ_{hi} to move.

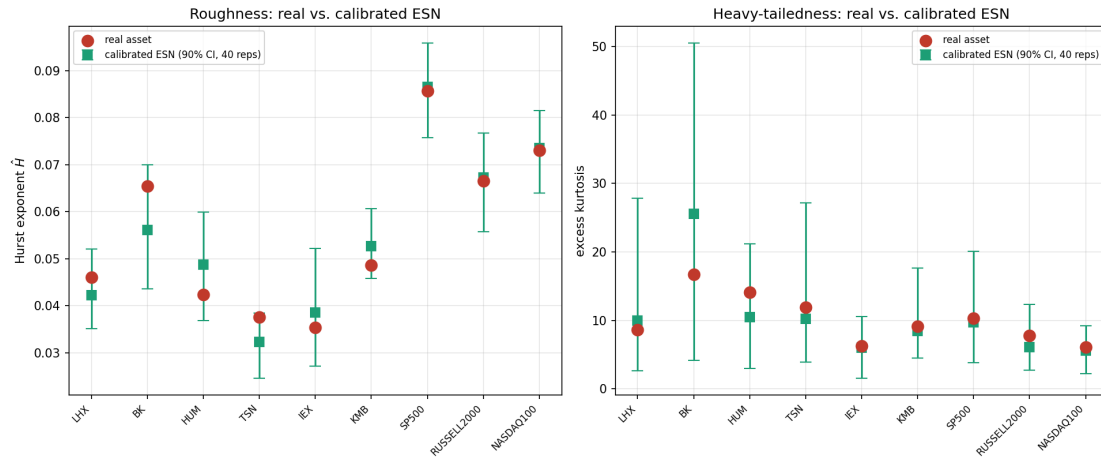


Figure 1: Roughness ($\hat{H}$, left) and excess kurtosis (right): real asset (red) vs. the jointly-calibrated ESN's own mean and 90% confidence band (green, 40 replicate $T = 4,000$ -day paths), all 9 assets. Both essentially perfect.

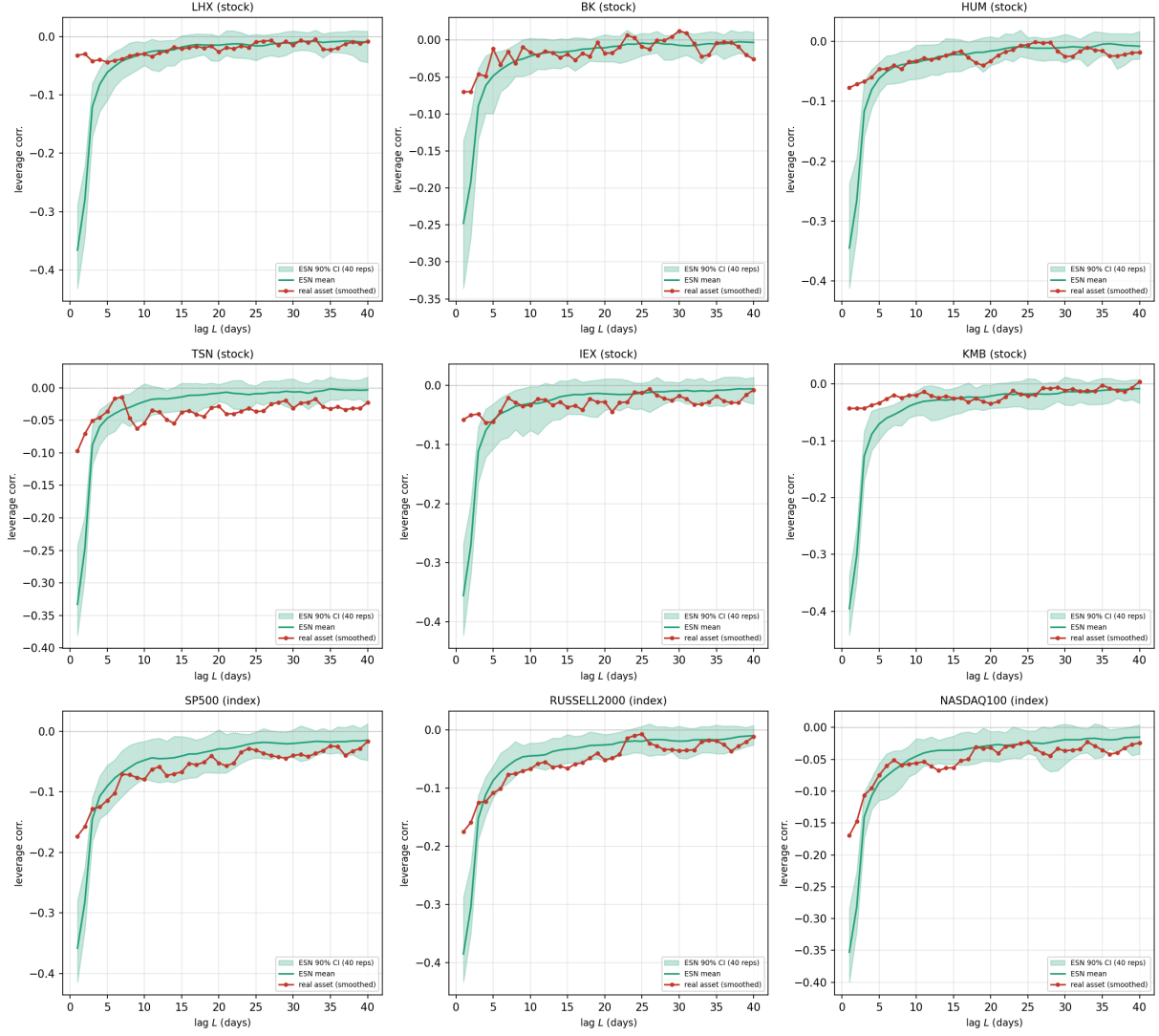


Figure 2: Leverage correlation vs. lag, all 9 assets – unaffected by the feedback layer, remaining excellent throughout.

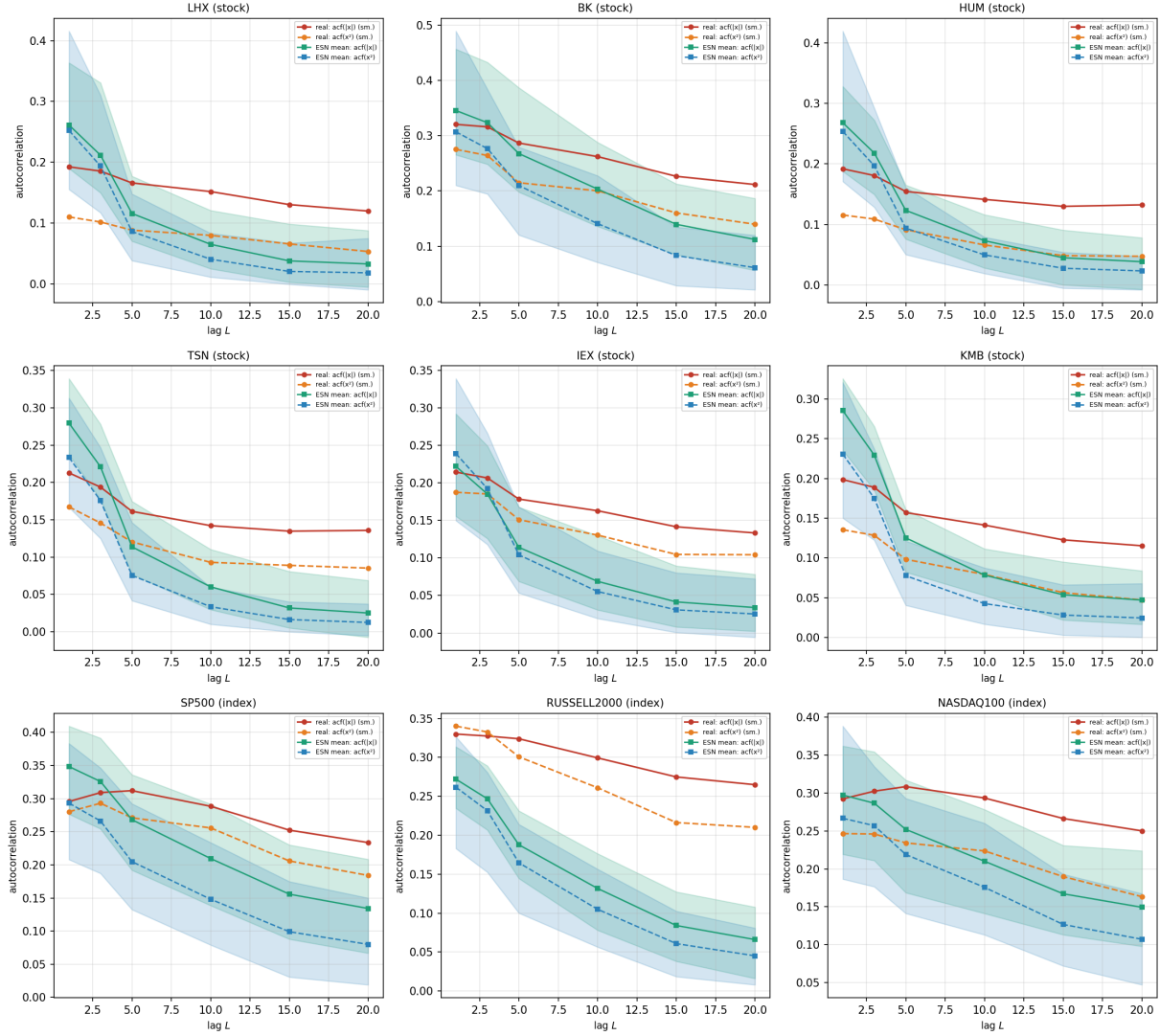


Figure 3: Autocorrelation of $|x_t|$ and x_t^2 vs. lag, all 9 assets. BK, TSN, KMB, SP500, NASDAQ100 all show the ESN's own mean tracking real data far more closely than the single-signal version's own collapse (cf. the previous version of this note) – only RUSSELL2000 remains at a strict 0% floor.

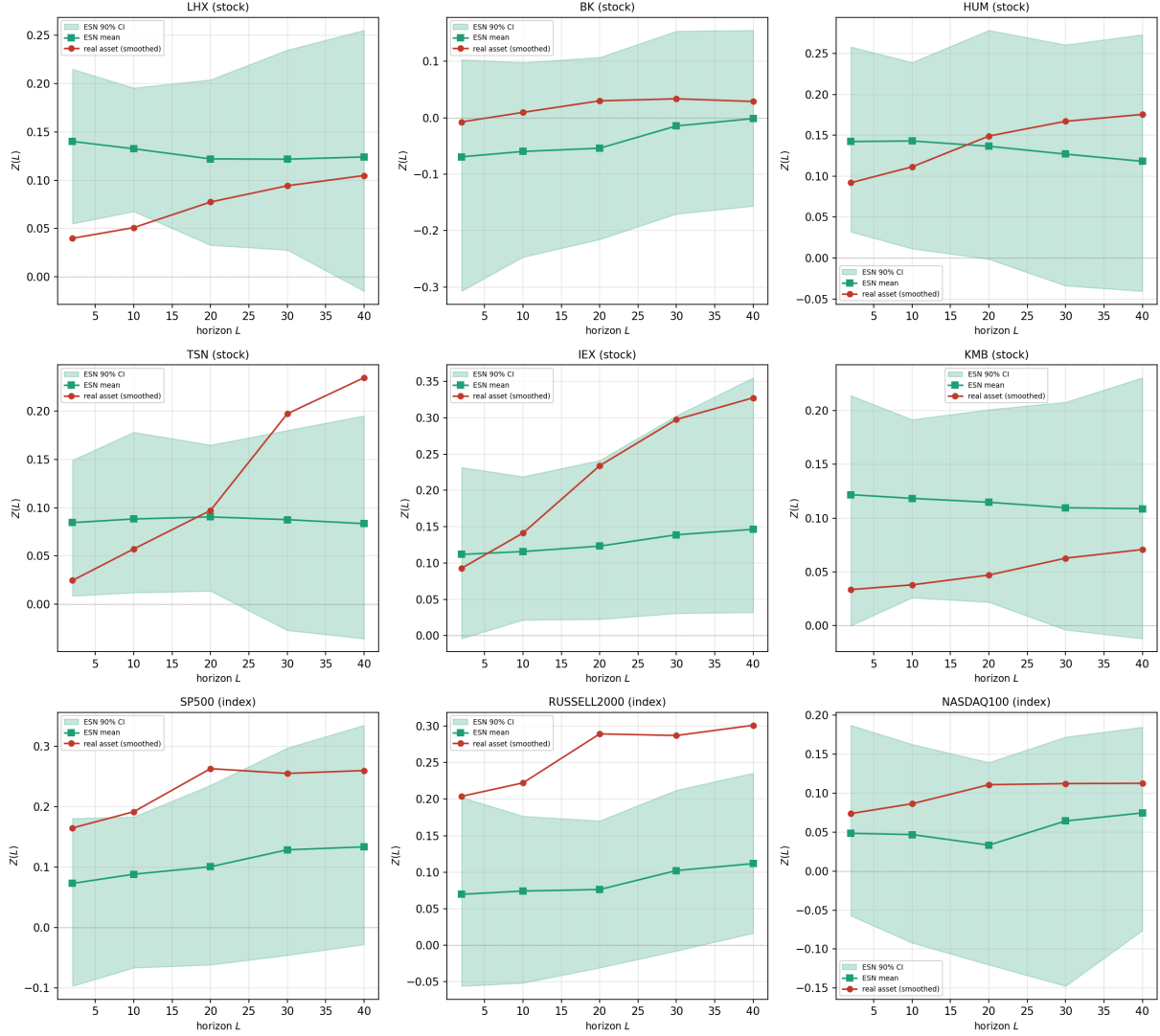


Figure 4: $Z(L)$ vs. horizon, all 9 assets. BK, HUM, IEX, KMB, SP500, RUSSELL2000, and NASDAQ100 – 7 of 9 assets – now show the ESN’s own mean genuinely rising with horizon, matching real data’s own qualitative shape, even where the magnitude still undershoots for several; only LHX and TSN stay flat against a clearly rising real curve.

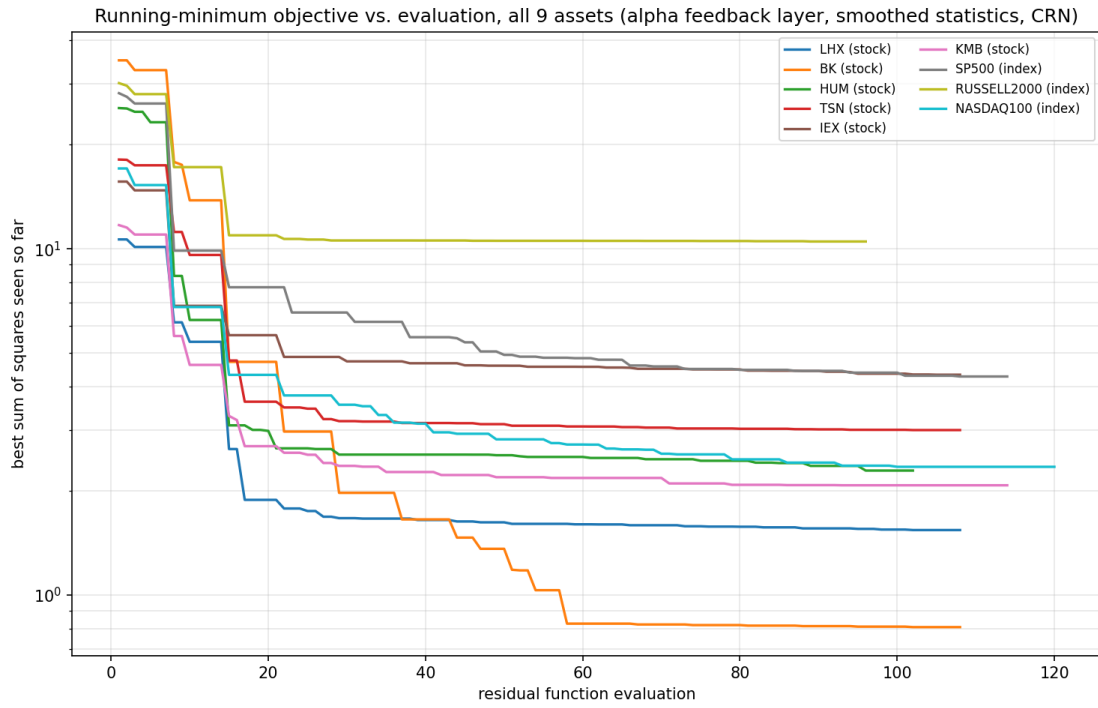


Figure 5: Running-minimum combined objective vs. residual-function evaluation, all 9 assets overlaid. Every asset shows a genuine, monotonic staircase decay under common random numbers (§3.5).

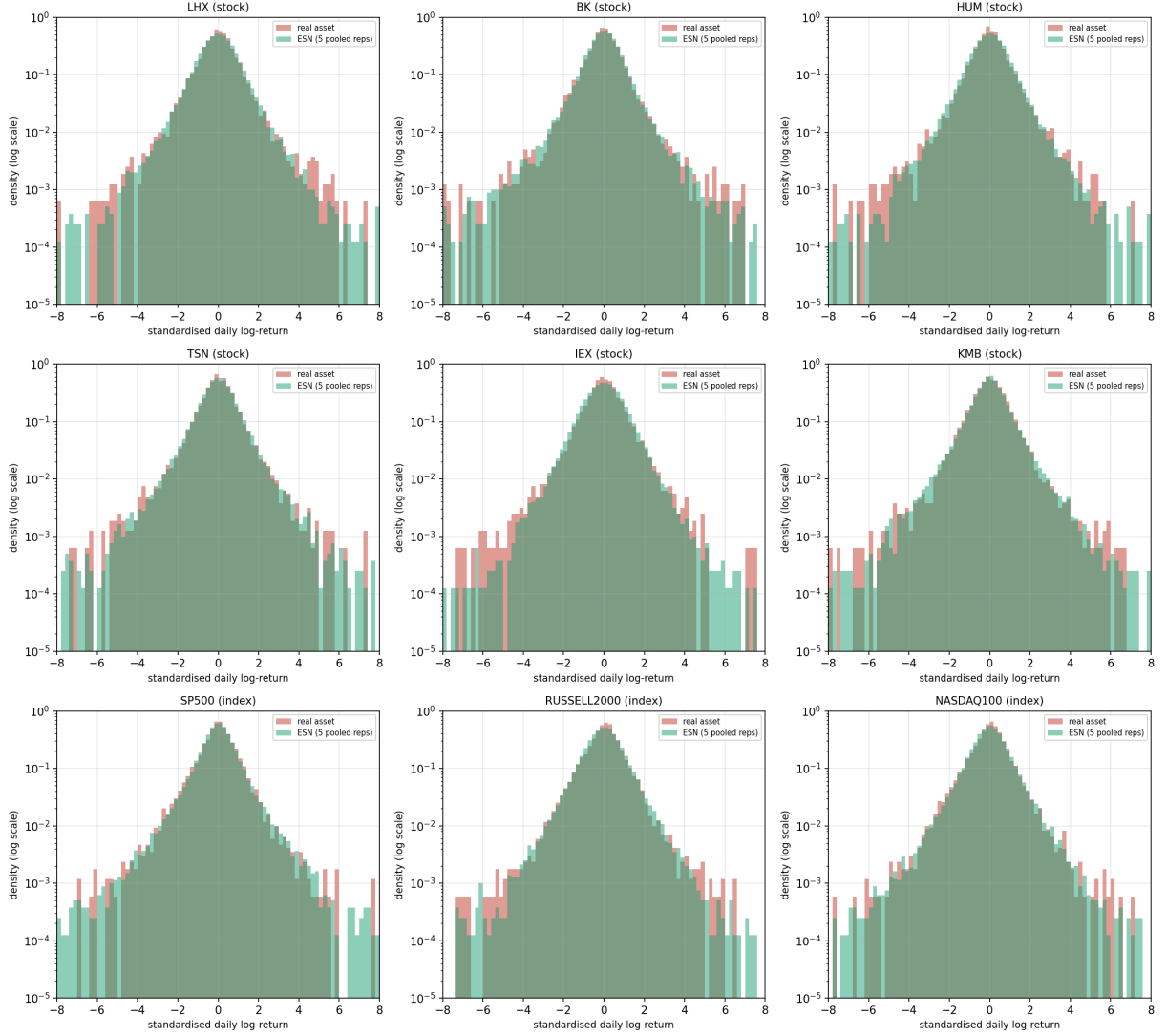


Figure 6: Empirical distribution of standardised daily log-returns, real asset vs. the calibrated ESN (5 pooled $T = 8,000$ -day replicate paths per asset, Table 1's own parameters), all 9 assets. Both series are standardised (mean-subtracted, std-normalised) before histogramming so shapes, not levels, are compared; the y -axis uses a log scale to make tail behaviour visible. The bulk of the distribution (roughly $|z| < 4$) matches closely for every asset – a direct visual confirmation of Table 2's own perfect kurtosis coverage – while the extreme tail ($|z| > 5-6$) is noisier and occasionally sparser in the ESN's own pooled sample for several assets (LHX, TSN, IEX particularly), consistent with kurtosis being a 4th-moment average that constrains the bulk of the distribution's own shape without pinning down the rarest, most extreme tail events.

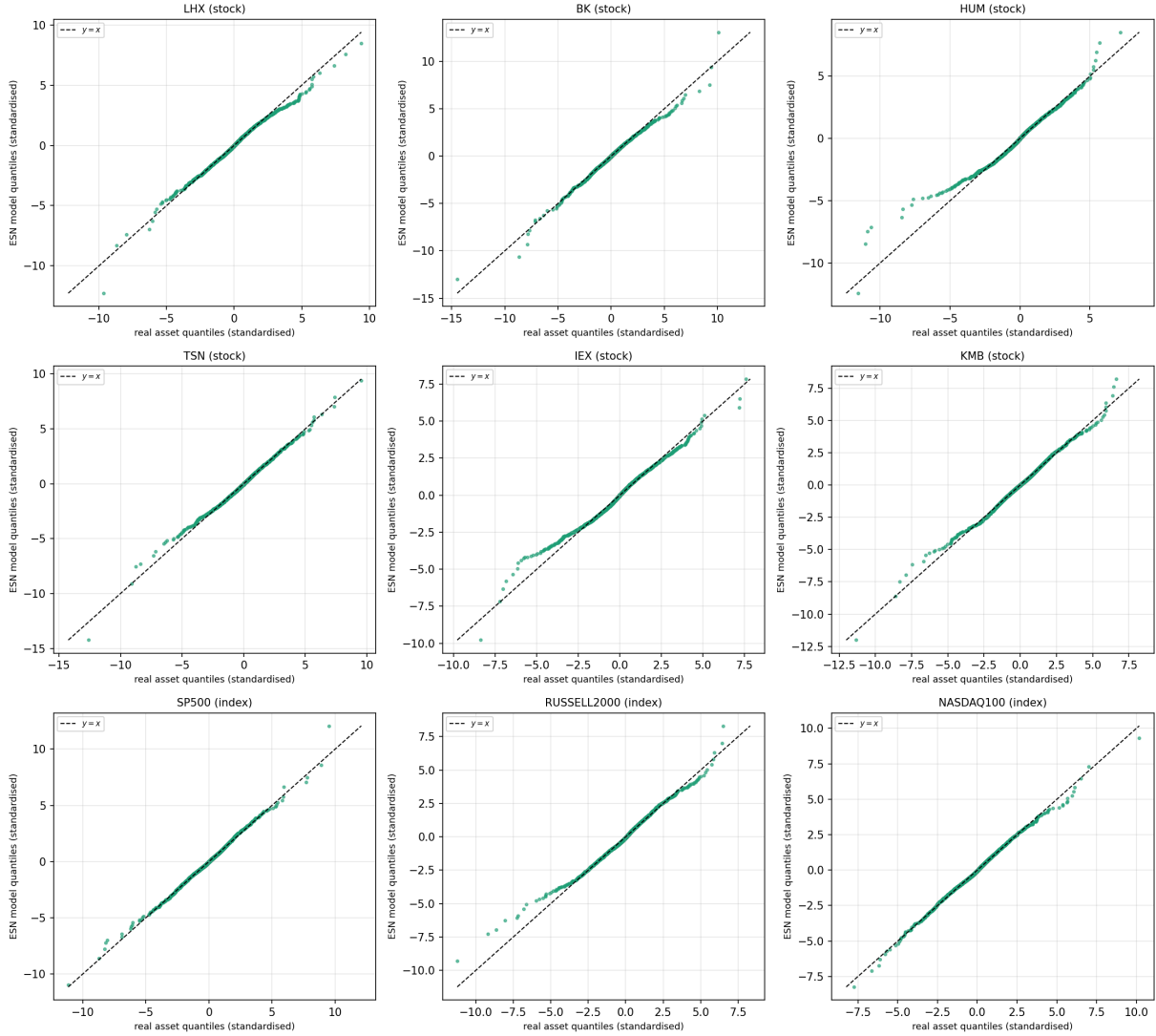


Figure 7: Q–Q plot of standardised daily log-returns, real asset quantiles (horizontal) vs. the calibrated ESN’s own quantiles (vertical, same 5 pooled replicate paths as Figure 6), all 9 assets; the dashed line is $y = x$. The bulk of the distribution (roughly $|z| < 3$) sits almost exactly on $y = x$ for every asset, a finer-resolution confirmation of Figure 6’s own bulk match. The tails reveal a pattern the histogram does not: for LHX, BK, HUM, TSN, and KMB, the ESN’s own negative tail falls consistently *below* $y = x$ – the model produces more extreme downside moves than real data at the same quantile. RUSSELL2000 shows the opposite pattern, with real data’s own negative tail more extreme than the model’s. SP500 and NASDAQ100 track $y = x$ closely into both tails with only minor deviation.

Table 2: Coverage: fraction of lags/horizons (or Y/N for the 2 scalar facts) where the real target falls inside the jointly-calibrated ESN’s own 90% confidence band (40 replicates), all 9 assets.

Asset	type	Hhat	leverage	acf(x)	acf(x^2)	Zumbach	kurtosis
LHX	stock	Y	90%	50%	67%	60%	Y
BK	stock	Y	80%	67%	67%	100%	Y
HUM	stock	Y	92%	50%	50%	100%	Y
TSN	stock	Y	38%	33%	50%	60%	Y
IEX	stock	Y	82%	33%	50%	100%	Y
KMB	stock	Y	80%	33%	83%	100%	Y
SP500	index	Y	85%	67%	50%	60%	Y
RUSSELL2000	index	Y	57%	0%	0%	0%	Y
NASDAQ100	index	Y	78%	50%	100%	100%	Y

Roughness: perfect, 9/9. **Kurtosis:** perfect, 9/9. **Leverage:** excellent for 8 of 9 assets (57–92%), unaffected by the feedback layer. **Taylor effect:** recovers substantially – 33–100% for 8 of 9 assets, up from a strict 0–17% floor for every asset in the pure single-signal version. **Zumbach effect:** 60–100% for 7 of 9 assets, with the qualitative shape now correct for the same 7 (Figure 4). **RUSSELL2000 is the clear exception**, at 0% on both Taylor curves and Zumbach despite 57% leverage coverage and perfect roughness/kurtosis.

5 Limitations and Recommended Next Steps

1. **RUSSELL2000 regresses sharply under this combination**, collapsing to 0% coverage on both Taylor curves and Zumbach simultaneously. Its own converged $\alpha = 0.229$ is not noticeably more extreme than several assets that succeeded (e.g. SP500’s own 0.220), so the cause is not obviously α itself; whether a different starting point resolves this was not tested.
2. **Even where Zumbach’s own shape is now correct (Figure 4), the magnitude frequently undershoots** (IEX, RUSSELL2000 particularly, where the ESN’s own mean rises much less steeply than real data). A dedicated per-asset amplitude correction was not attempted.
3. **The 3-point smoothing window (§3.2) and the feedback layer’s own starting value ($\alpha_0 = 0.1$) were not themselves tuned**; different choices were not tested against these.
4. **Common random numbers (§3.5) tie the optimiser’s own trajectory to 1 specific noise realisation per asset**; the final confirmation step uses a separate, fixed seed (888) independent of the optimisation trajectory, but the stage-1 search itself optimises against 1 realisation, not the true expectation.
5. **m_2 and the z -bank’s own timescale range and readout profile remain fixed throughout**; a joint search over a wider parameter set was not attempted.